Experiment - 8.1.2. Prime Numbers in a Given Range

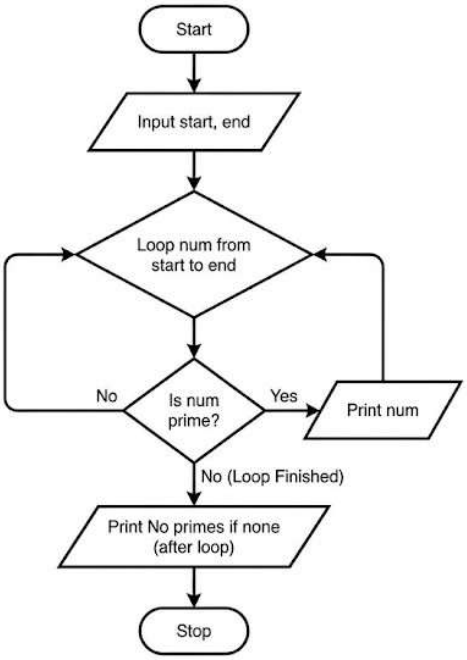
1. Aim

To design and implement a Python program that finds and prints all prime numbers within a specified range [start, end] (inclusive). The program determines the primality of each number and prints it. If no prime numbers exist within the given range, the program prints "No primes".

2. Pseudocode

1. **START**
2. **READ** an integer representing the start of the range and **STORE** it in start.
3. **READ** an integer representing the end of the range and **STORE** it in end.
4. **INITIALIZE** a boolean variable found to False to keep track of whether any primes exist.
5. **FOR** each number num in the range from start to end (inclusive):
    - **IF** num > 1 (since 1 and numbers below it are not prime):
        - **INITIALIZE** is_prime to True.
        - **FOR** each number i from 2 up to the integer square root of num (int(num**0.5) + 1):
            - **IF** num is divisible by i (num % i == 0):
                - **SET** is_prime to False.
                - **BREAK** the inner loop.
        - **IF** is_prime is True:
            - **PRINT** the prime number num.
            - **SET** found to True.
6. **IF** found is False (meaning no primes were printed):
    - **PRINT** "No primes".
7. **END**

3. Flowchart



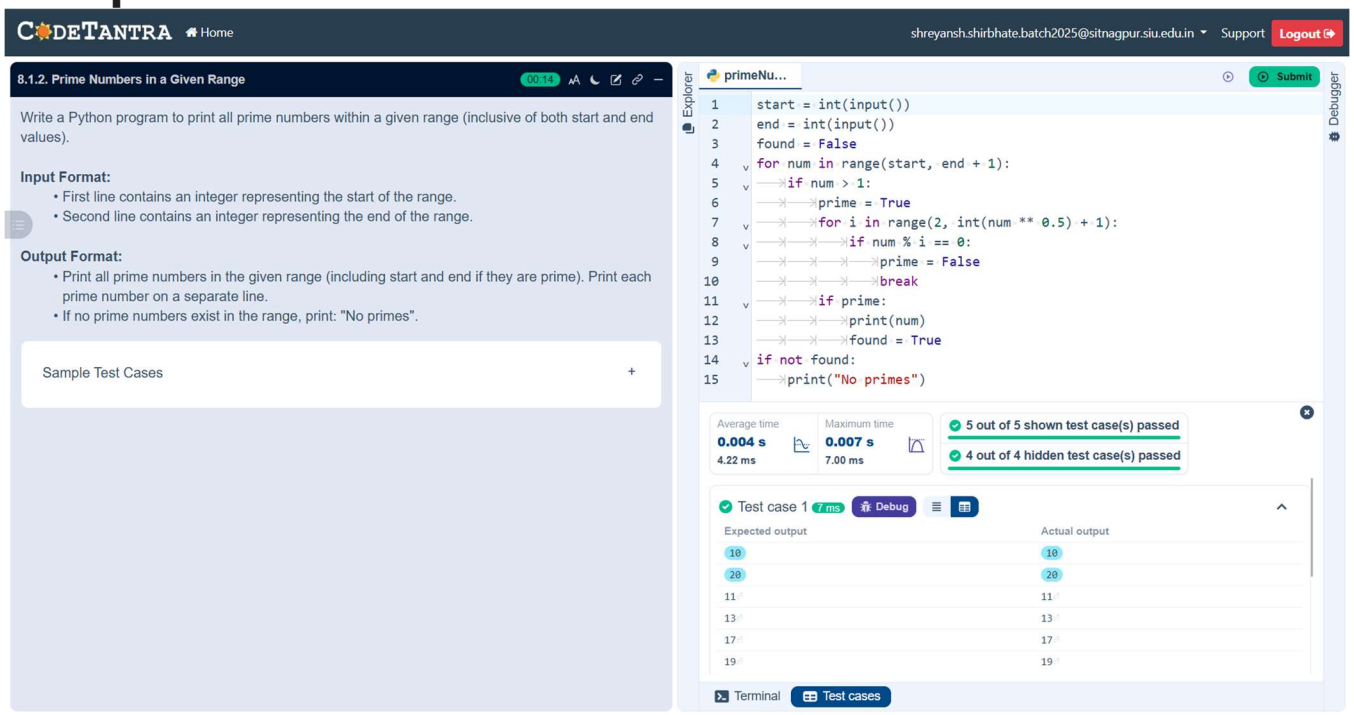
4. Python Program

```
start = int(input())
end = int(input())

found = False

for num in range(start, end + 1):
    if num > 1:
        is_prime = True
        for i in range(2, int(num**0.5) + 1):
            if num % i == 0:
                is_prime = False
                break

        if is_prime:
            print(num)
            found = True

if not found:
    print("No primes")
```

5. Experiment Screenshot